# American Computer Science League

2023-2024 ● Contest 3: Short Problems Solutions ● Senior Division

**1. Boolean Algebra**

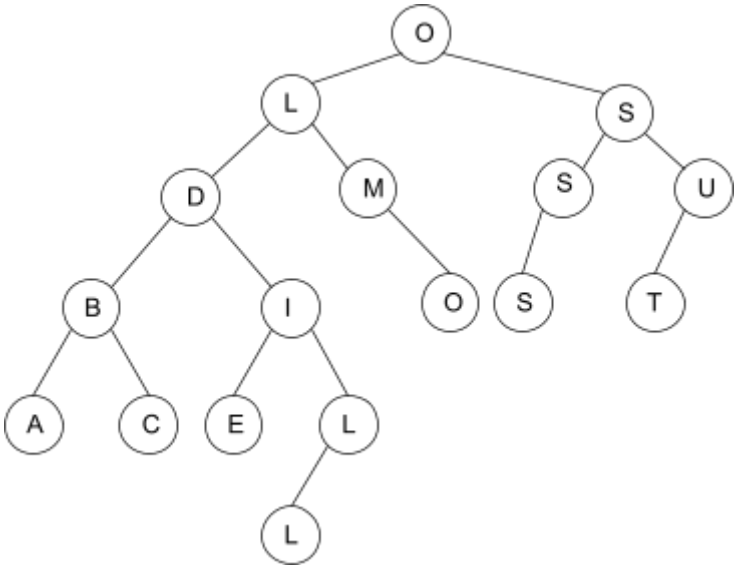
$\overline{A(\overline{B\,C\ +\ \overline{A}})}\,\overline{(\overline{A}\,B\,C)}$
$= A\,\overline{\overline{BC}}\,\overline{\overline{A}}\,(\overline{\overline{A}\ +\ \overline{B}\ +\ \overline{C}})$
$= A\,(\overline{B}\ +\ \overline{C})\,(A\ +\ \overline{B}\ +\ \overline{C})$
$= (A\,\overline{B}\ +\ A\,\overline{C})\,(A\ +\ \overline{B}\ +\ \overline{C})$
$= A\,\overline{B}\,A\ +\ A\,\overline{B}\,\overline{B}\ +\ A\,\overline{B}\,\overline{C}\ +\ A\,\overline{C}\,A\ +\ A\,\overline{C}\,\overline{B}\ +\ A\,\overline{C}\,\overline{C}$
$= A\,\overline{B}\ +\ A\,\overline{B}\ +\ A\,\overline{B}\,\overline{C}\ +\ A\overline{C}\ +\ A\,\overline{C}\,\overline{B}\ +\ A\,\overline{C}$
$= A\,\overline{B}\ +\ A\,\overline{B}\,\overline{C}\ +\ A\,\overline{C}\ +\ A\,\overline{C}\,\overline{B}$
$= A\,\overline{B}\,(1\ +\ \overline{C}\ +\ \overline{C})\ +\ A\,\overline{C}$
$= A\,\overline{B}\ +\ A\,\overline{C}$
$= A(\overline{B}\ +\ \overline{C})$

To be TRUE $A\ =\ 1\ and\ (\overline{B}\ +\ \overline{C})\ =\ 1$.
Therefore, the ordered triples are: (1,1,0), (1,0,1), (1,0,0)

B. 3

**2. Data Structures**

The binary search tree for OLDSMOBILECUTLASS is:




The internal path length is: 2*1 + 4*2 + 5*3 + 4*4 + 5*1
= 2 + 8 + 15 + 16 + 5
= 46

C. 46

**3. Data Structures**

Booth 1 - Amy (15) + Gigi (2/17) + Hal (12/29) + Lexi (8/37) + Omar (5/42)
Booth 2 - Beth (10) + Eve (4/14) + Fran (7/21) + Jan (12/33) + Max (8/41)
Booth 3 - Cam (8) + Deb (11/19) + Ivy (9/28) + Kay (6/34) + Nara (10/44)

C. Booth C - 44

**4. FSAs**

a. 1101110 - top path - fails at the 2
b. 1121022 - accepted
c. 1111111 - accepted
d. 1010 - accepted
e. 1102112 - top path - fails at final 0
f. 1011112 - bottom path - fails at final 0

D. b, c, d

**5. Regular Expressions**

The following do not satisfy the given regular expression.

[a-p]* e* [^s]* [d-hp-y]

1. relish fails at h in [d-h, p-y]
2. salt fails at a in [d-h, p-y]
3. mustard fails at t in [d-h,p-y]
4. mayo fails at o in [d-h, p-y]
5. soy fails at o in [d-h, p-y]
6. salsa fails at a in [d-h, p-y]

Therefore, 6 (pickle, pepper, ketchup, butter, vinegar, honey) are valid and 6 fail.

C. 6